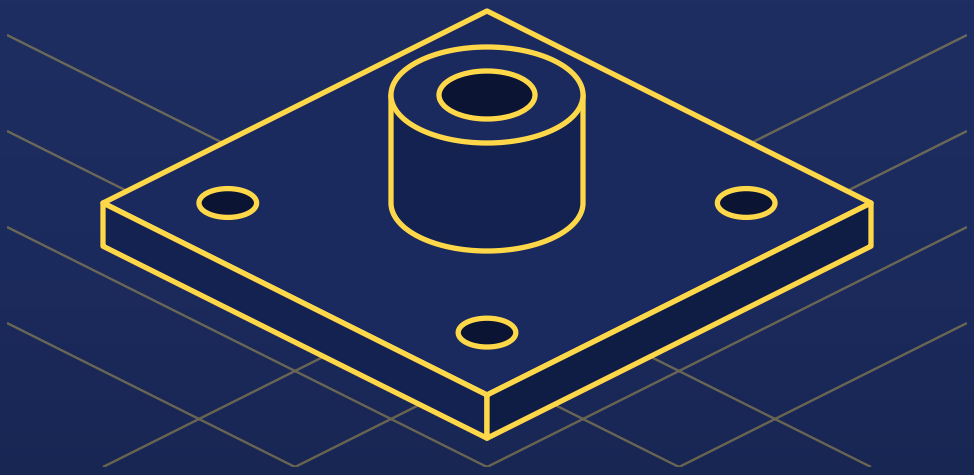
SolidCraft 98

The Definitive Guide

Parametric solid modelling, one rectangle at a time

Covers Turbo!



- Sketch, extrude, revolve and drill
- Change your mind afterwards
- Export for Rapid Prototyping
- What the Turbo button really does

BACK OFFICE PRESS

No CD-ROM included. It fits on a floppy.

About this book

SolidCraft 98 is a parametric CAD program for the desktop PC. You draw a flat shape, pull it into a solid, and drill holes in it. Then you change any number you like, and the part follows.

This book is short on purpose. It shows you the screen, walks you through one real part from start to finish, and then lets you get on with it. Every picture is a real screenshot, taken on ReactOS at 800 × 600, which is the resolution SolidCraft 98 was meant for.

You need: a PC, SolidCraft 98 1.1.1, and about ten minutes.

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2. A tour of the screen
3. Your first part
4. Changing your mind
5. Shading it, and printing it
6. The Turbo button
7. Things to try
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CHAPTER 1

Getting it running

Download `SolidCraft98-1.1.1.zip` from the Releases page, unzip it anywhere, and run `SolidCraft98.exe`. There's no installer and no setup program. (There is a setup program, but it's not for installing. See Chapter 7.)

It runs on Windows 10 and 11 and on ReactOS. If you'd rather have it on a floppy, each release also has a 1.44 MB disk image.

> **Tip.** On Linux or a Raspberry Pi, get the Delphi Edition or the FreeBASIC version from the same Releases page. Everything in this book works the same way in them, and the files are the same files.

CHAPTER 2

A tour of the screen

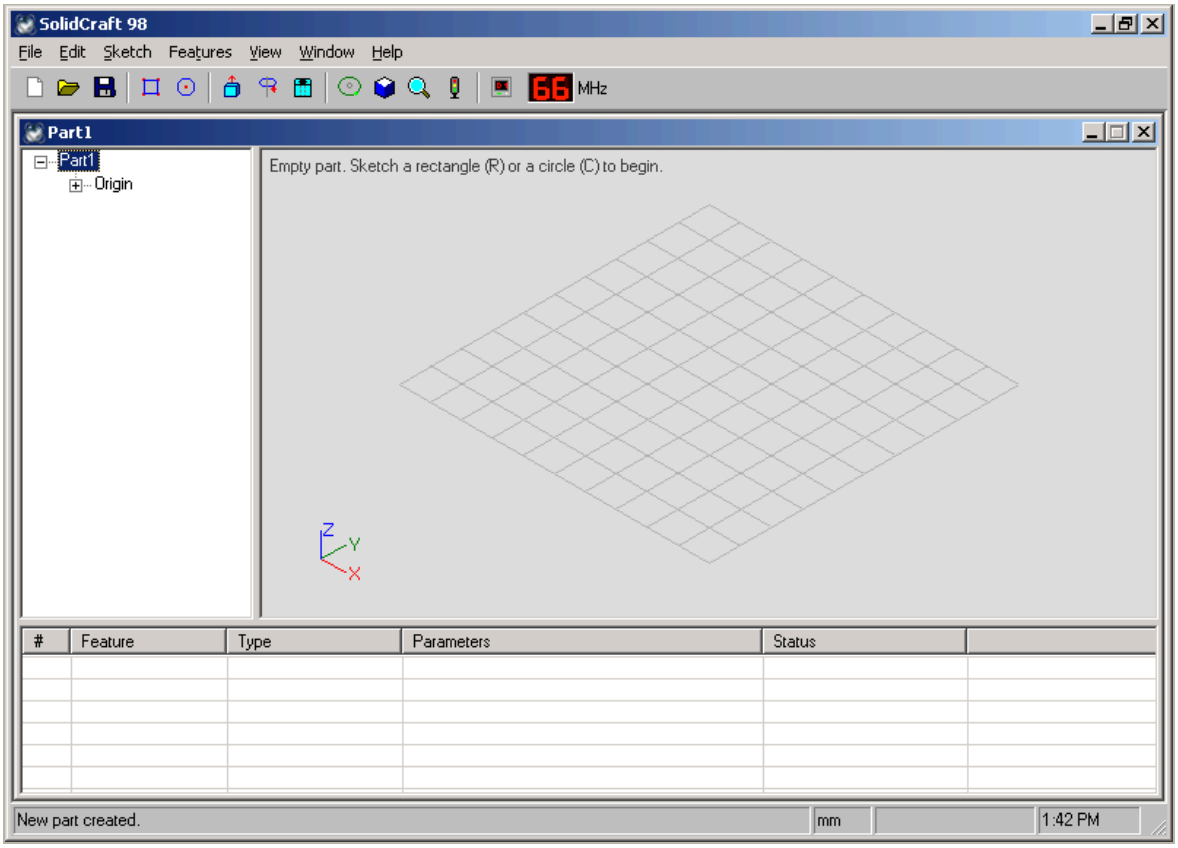


Figure 2-1. A new part. Nothing in it yet but a grid and some hope.

The toolbar

New, Open and Save; Sketch Rectangle and Sketch Circle; Extrude, Revolve and Hole; the views; and, on the right, the Turbo button and its LED. Point at a button to see what it does.

The feature tree

On the left. Everything you make goes in here, in the order you made it.

The view

Your part, from the front, the right and above. Drag with the left mouse button to turn it round, with the right button to slide it about, and press `+` or `-` to zoom.

The timeline

Along the bottom: every feature with its numbers. This is where you change your mind.

The status bar

What just happened, the units (always mm), and the time.

CHAPTER 3

Your first part

We'll make a mounting plate: 80 × 50 × 6 mm with round corners, a hole in each corner, and a boss in the middle with a hole through it. You'll find the finished one in `Samples\Mounting Plate.sc98` if you'd like to peek.

The outline

Press `R` for Sketch Rectangle. Every command asks for its numbers in the same kind of box. Type a number, press `Tab` for the next one, and `Enter` when you're done.

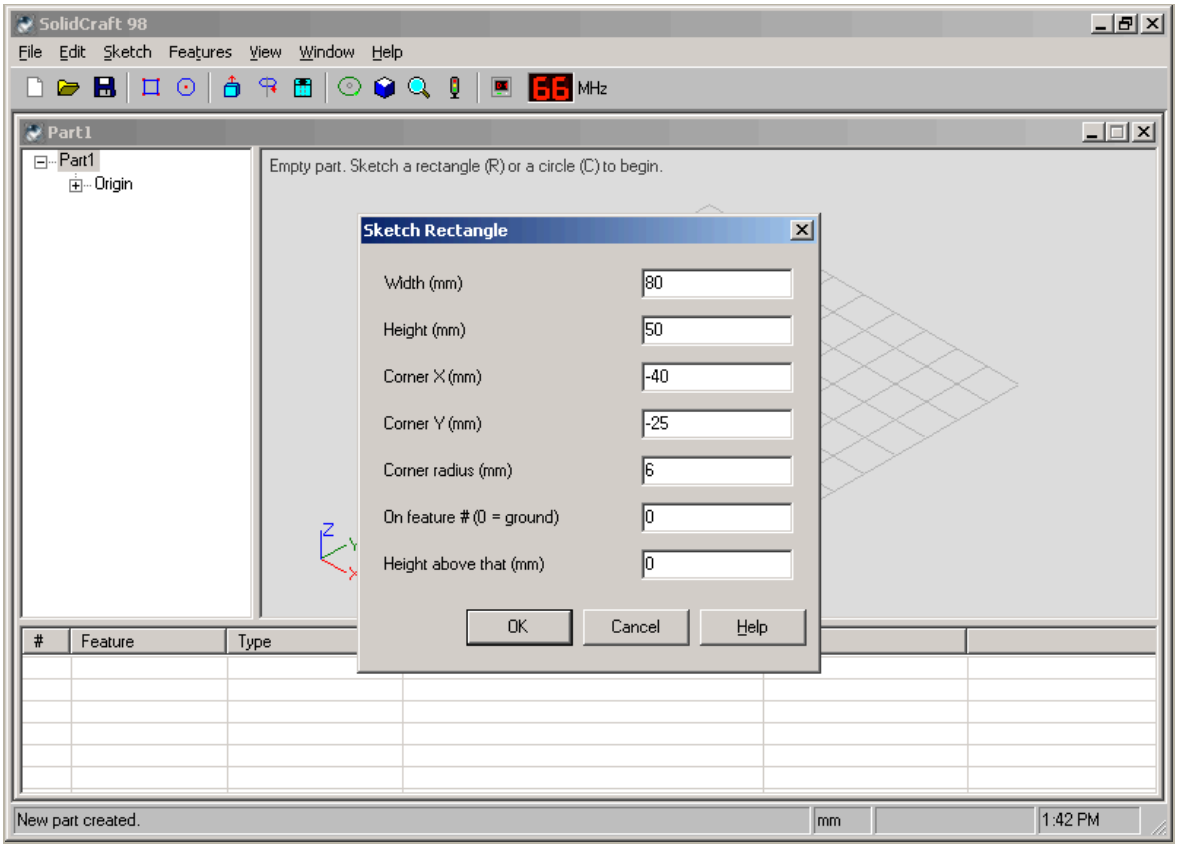


Figure 3-1. Width 80, Height 50, Corner X −40, Corner Y −25, Corner radius 6. Leave the rest at 0: the ground.

> **Tip.** Number boxes do arithmetic. `80/2` is 40, and `25+3.5` is 28.5. They don't do units, and they'll tell you so.

Meet Cal

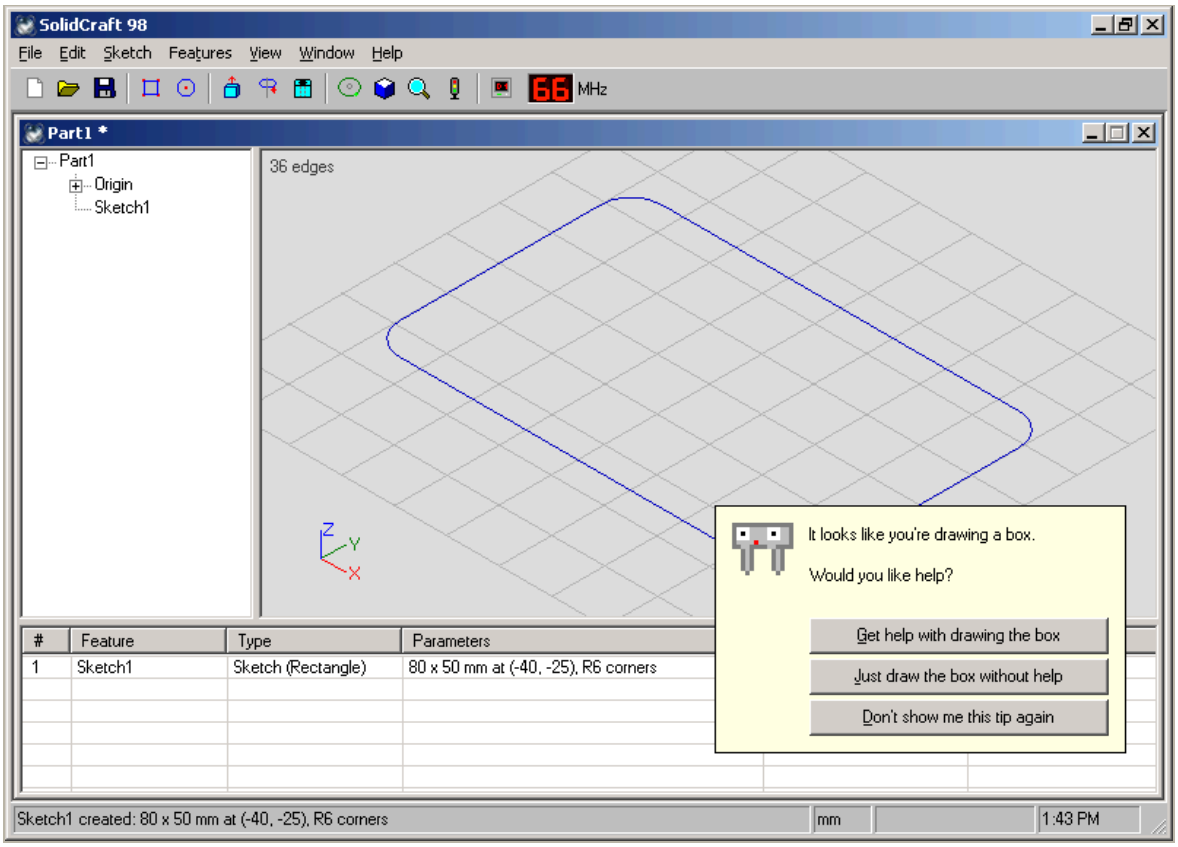


Figure 3-2. Cal, the caliper assistant, has noticed you're drawing a box.

Cal pops up the first time you sketch a rectangle. Cal means well. Pick whichever button you like; your rectangle is drawn either way.

The plate

Press `E` for Extrude, type 6, and press `Enter`. The outline becomes a solid plate, and Extrude1 appears in the tree and the timeline.

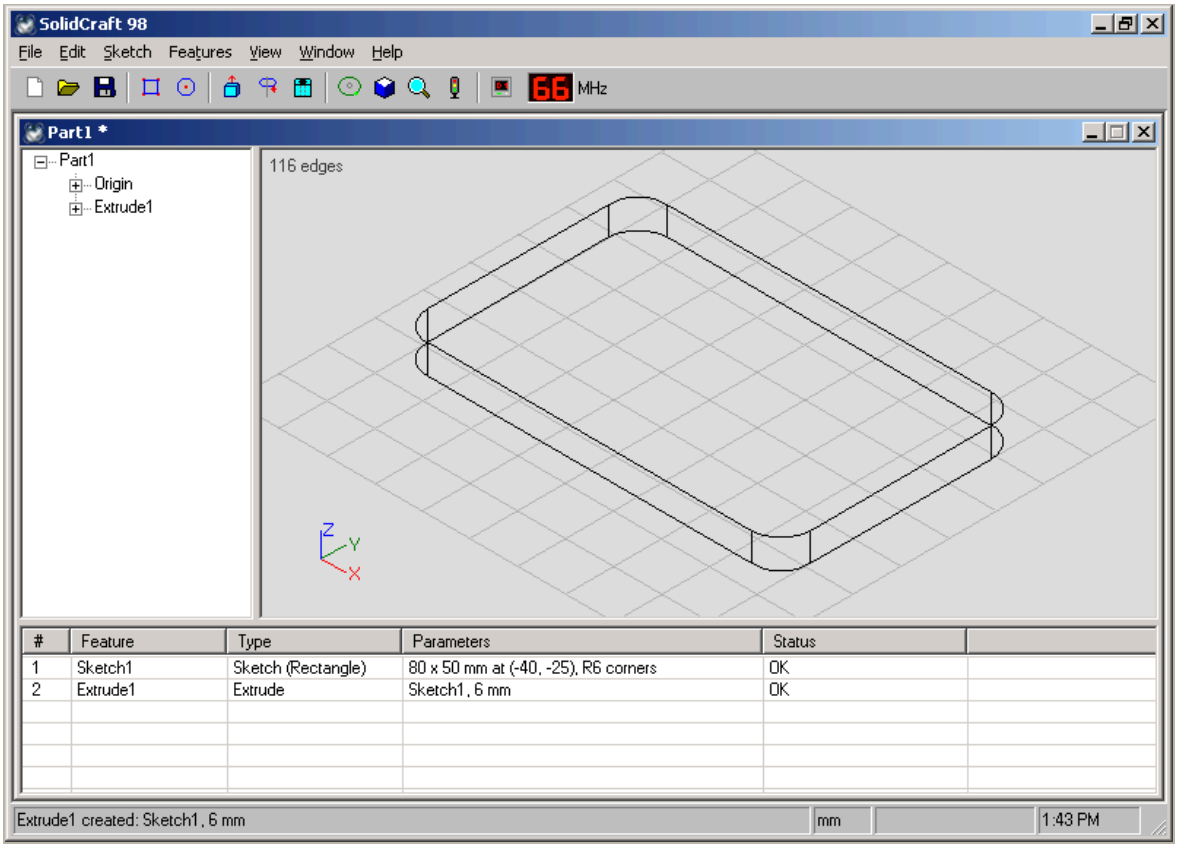


Figure 3-3. A plate. Already more than some people manage.

The holes

Press H for Hole. Diameter 5, Depth 0 (0 means all the way through), Centre X −32, Centre Y −17. Do it three more times at (32, −17), (32, 17) and (−32, 17).

The boss

Press C for Sketch Circle: Radius 10, Centre 0, 0. Set *On feature #* to 2, the plate's number in the timeline's # column, so the circle sits on top of the plate. Then E again, 15 mm. Last, one more H: Diameter 10, Depth 0, Centre 0, 0. It goes into the boss, and because it goes all the way through, it carries on through the plate too.

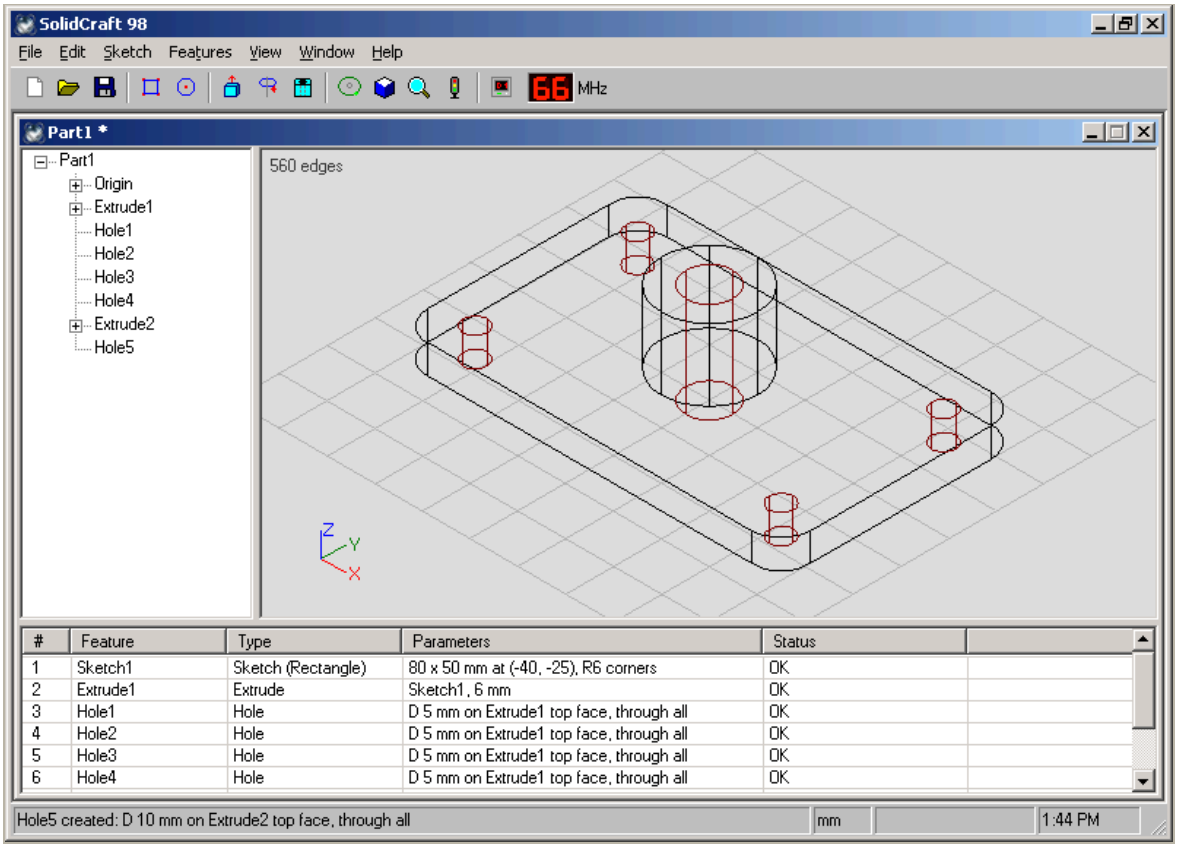


Figure 3-4. The finished part, and the timeline that made it.

Save it with `Ctrl` + `S`. Parts save as `.sc98` files, which only SolidCraft 98 can read. This is traditional.

CHAPTER 4

Changing your mind

Here's the clever bit. Every feature remembers its numbers, and everything built on it follows when they change.

Double-click **Extrude1** in the timeline (or select it and press `Enter`). Make the plate 8 mm instead of 6, and press `Enter`.

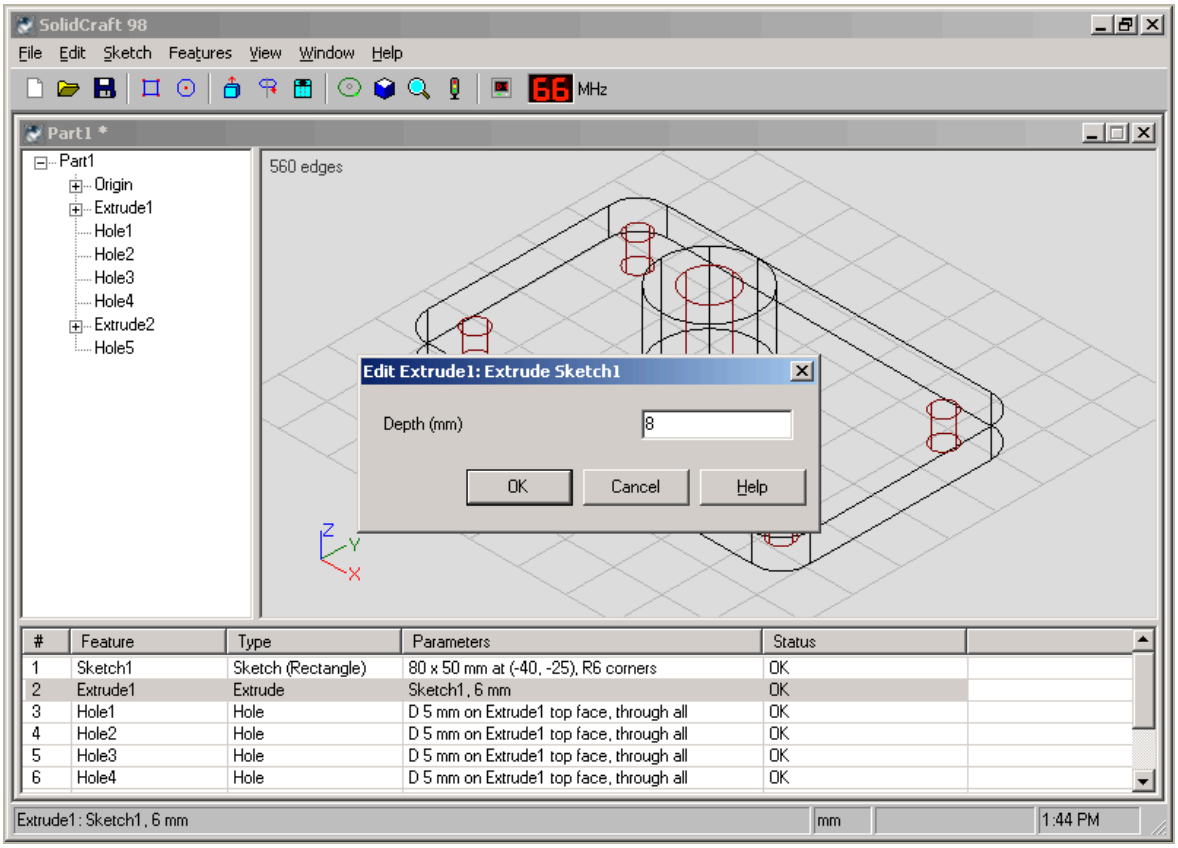


Figure 4-1. Editing Extrude1. The boss will move up with the plate, and the bore still goes all the way through.

Changed your mind about the change? `Ctrl` + `Z` takes off the last feature you added, and `Del` deletes the selected one. If anything was built on it, SolidCraft 98 asks first.

CHAPTER 5

Shading it, and printing it

Press S. SolidCraft 98 works out every pixel of every face itself, then shows the picture a few rows at a time, top to bottom, the way a good render should. Press S again for wireframe.

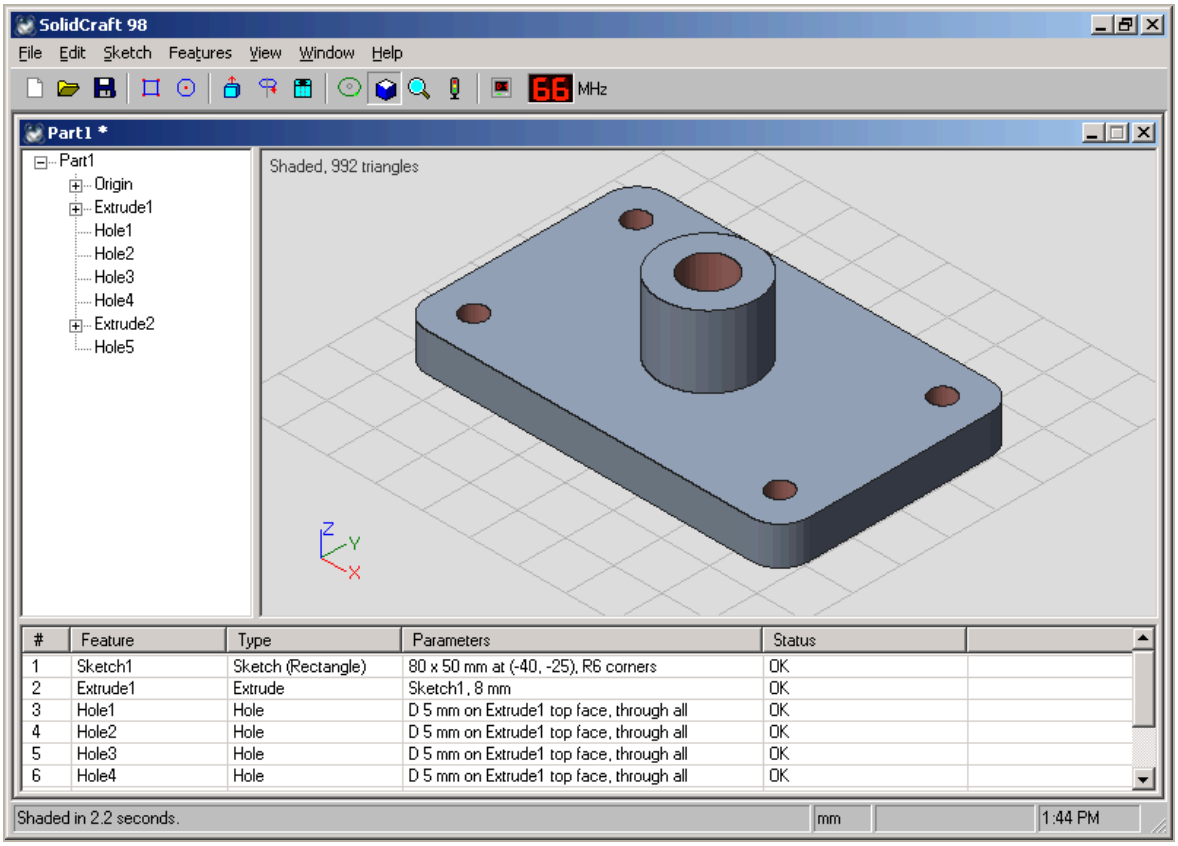


Figure 5-1. Shaded. It takes a second or two, on purpose.

To 3D-print it, choose **File › Export for Rapid Prototyping (*.stl)**. You get an STL file in millimetres that any slicer opens. Holes go all the way through, and the boss is joined to the plate, so it prints as one solid.

CHAPTER 6

The Turbo button



Figure 6-1. Turbo on. Note the LED.

Like the button on the front of a 1990s PC. Press it and the LED changes from 66 to 33. That's not a typo.

Turbo switches on the full 1999 experience. Some things get slower, some things get more talkative, and a few things you wouldn't expect. With Turbo off, everything just works.

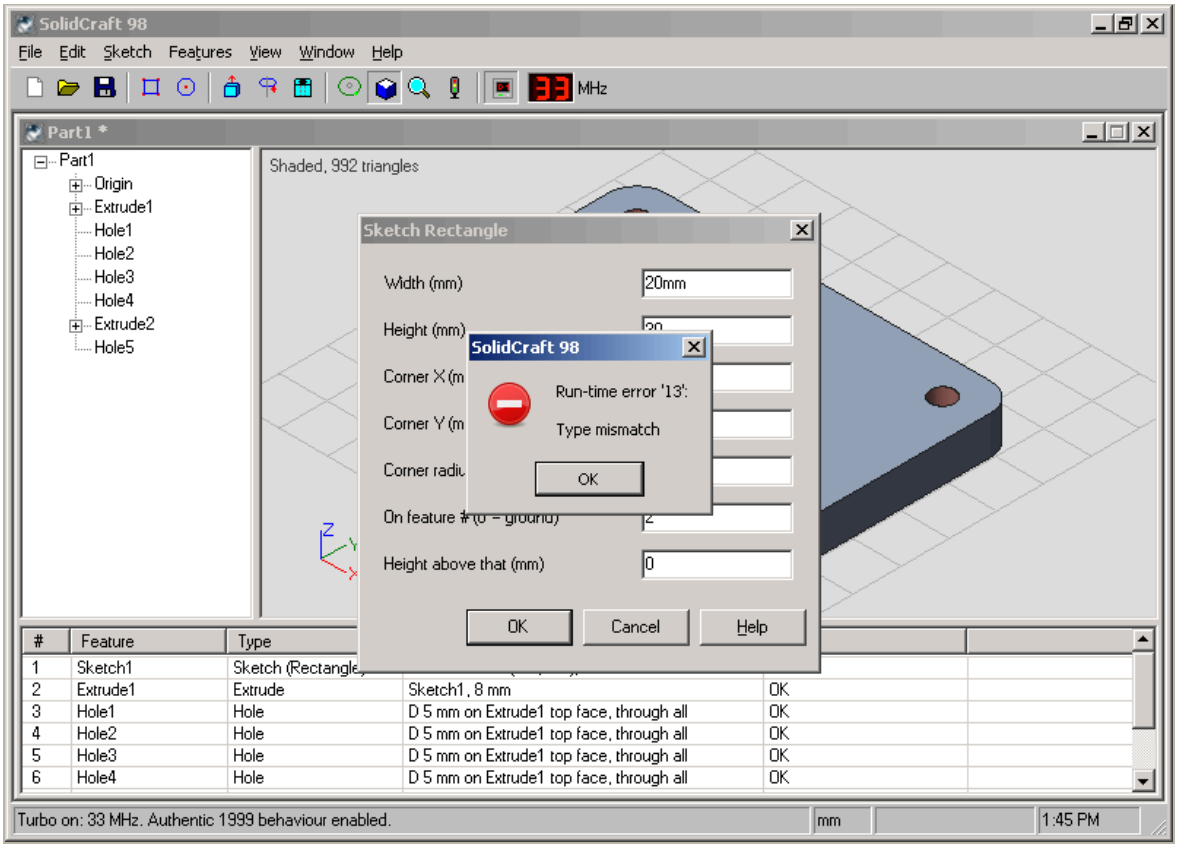


Figure 6-2. Typing `20mm` into a number box. Keep trying. It gets more personal.

> **Watch out.** Nothing Turbo does touches your files, your clock or your disk drives, and no part is ever lost. It only looks like it might.

CHAPTER 7

Things to try

No spoilers. Just hints.

- Open **Help › About** and type the name of a famous operating system.
- With Turbo on, Shift-click the logo in the About box. Then wait.
- Try **File › Send To 3½ Floppy (A:)**. Then try it with Turbo on.
- Look at your part, and enter the code every 1990s gamer knows: ↑ ↑ ↓ ↓ ← → ← → B A.
- Divide by zero in a number box, with Turbo on. Your part is fine. Promise.
- **File › Publish to Web**. Every part deserves a home page.
- Start it with `/Y2KTEST`, turn on Turbo, and wait a minute.

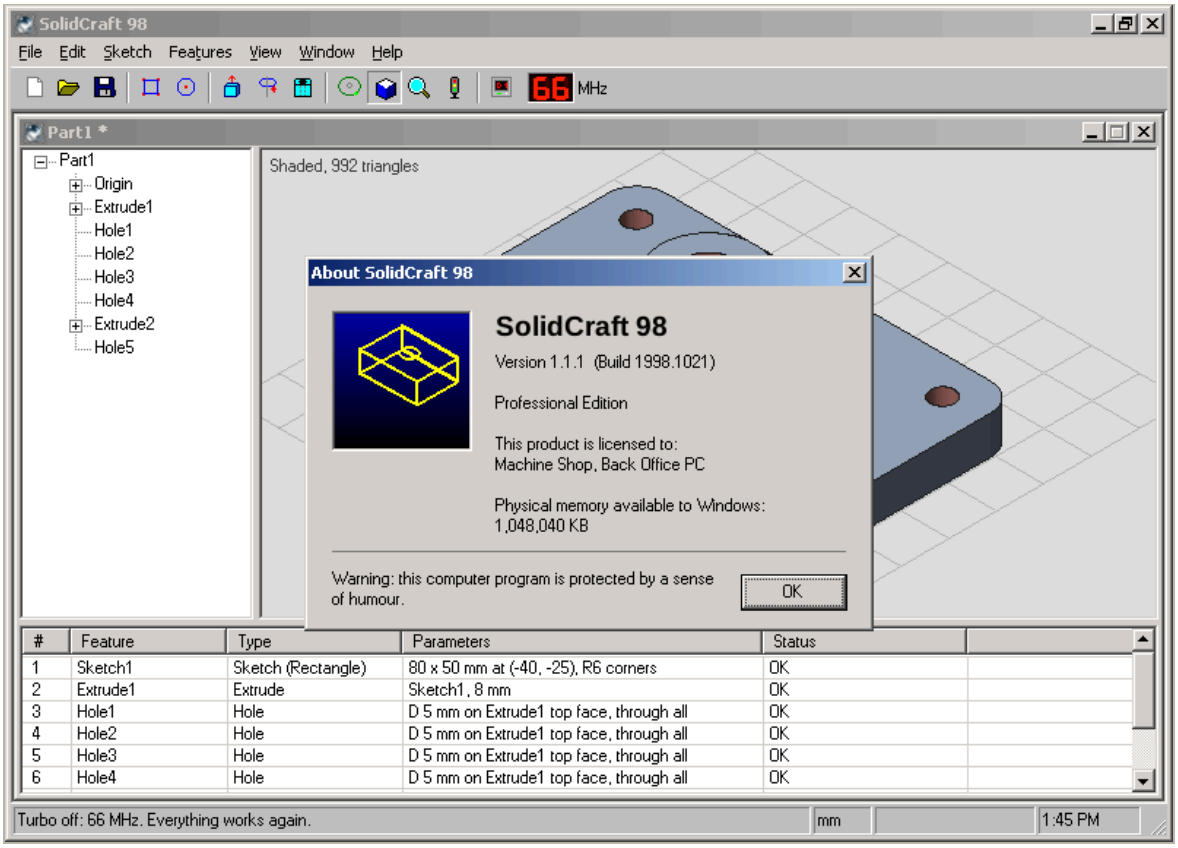


Figure 7-1. The About box. Start here.

APPENDIX

The keys

`Ctrl` + `N` / `O` / `S`	New, Open, Save
`Ctrl` + `Z`	Undo the last feature
`R`, `C`	Sketch a rectangle, a circle
`E`, `V`, `H`	Extrude, Revolve, Hole
`I`, `F`	Isometric view, Zoom to fit
`S`	Shaded (again for wireframe)
`Enter`, or double-click	Edit the selected feature
`Del`	Delete the selected feature
`+` / `-`	Zoom in, zoom out
`F5`	Regenerate
`F1`	Help (it isn't installed)

"It's a joke, but it works."

SolidCraft 98 is a parametric CAD program written the way it would have been in 1999. This is the book that came in the box. Well, it would have.

SolidCraft 98 is a parody. It isn't affiliated with, endorsed by, or a replacement for SolidCraft. Back Office Press is as real as the Turbo button. Text and pictures MIT licensed, like the program.